Ultimate Ontological Structures and the Demarcation of Epistemological Barriers: The Ultrastructural Limit of Discrete Cognition

Dmitry N. Boyko
Kemerovo, Russia
dmitry.boyko@gmail.com
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Abstract

In the present work, using methods of category theory, quantum information theory, and non-equilibrium thermodynamics, we rigorously prove the existence of an upper combinatorial limit of abstraction for discrete cognitive systems of the human type. The evolution of sign systems is formalised as an orthodoxly directed process of increasing relational topology complexity, from binary graphs to n-dimensional nested ultragraphs. An isomorphic correspondence is established between the algebraic structure of an ultragraph and the multidimensional geometry of the amplituhedron in the Grassmannian outside space-time. Based on the Bekenstein bound and entropy transport equations, the theoretical collapse of any higher-order metastructures into isotropic white noise is proved. The biochemical censorship of the carbon substrate is formulated as an irreducible limitation on the throughput capacity of the observer's codec.

1 Introduction: Axiomatics of Cognitive Quantization

Within the paradigm of interface theory of perception [2], the space-time continuum $\mathcal{M}$ is defined not as a fundamental ontological given, but as an emergent user interface — an evolutionary compression codec mapping an infinite-dimensional information space into a low-dimensional representation. The cognition of reality by a discrete observer $\mathcal{O}$ is equivalent to the continuous application of a selection (discretisation) operator.

We define an elementary cognitive act as a mapping Ψ that translates a continuous isotropic field of quantum amplitudes $\mathcal{H}$ into a discrete semantic subset:

$$\Psi : \mathcal{H} \longrightarrow \mathcal{S} = \{V, E\} \tag{1}$$

where V is a set of distinguished entities (vertices) and E is a set of binary relations between them (edges). This process is isomorphic to a classical quantum measurement: the reduction of the wave function under decoherence produces a discrete fact and fixes its relation to the measurement basis.

The historical trend of increasing complexity of mathematical abstractions is expressed in the successive transition from simple binary structures to structures with intersecting higher-order contexts. The present study aims to rigorously prove that the pair "Ultragraph – Amplituhedron" constitutes an absolute and final epistemological barrier (the "Wall of Cognition") for discrete systems of the human type.

2 Algebraic-Geometric Isomorphism: Ultragraph and Amplituhedron

2.1 Categorical Formalism of Ultrastructures

We formulate an Ultragraph as a generalised combinatorial structure that abolishes the rigid stratification between elements of sets. Let $X_0 = V$ be the initial set of vertices. We inductively define the family of higher-order sets:

$$X_{k+1} = X_k \cup \mathcal{P}(X_k), \quad k \in \mathbb{N}_0 \tag{2}$$

Then the Ultragraph $\mathcal{U}$ is the ordered pair:

$$\mathcal{U} = (V, E), \quad E \subseteq \prod_{k=0}^{\infty} X_k \tag{3}$$

In categorical terms, an ultragraph represents a free monoidal category where objects and morphisms are recursively embedded, forming a topology of the "set of all subsets" type. The ultragraph is the ultimate algebraic expression of contextuality: a higher-order edge $e_i \in E$ can serve as a context (modifier) for a lower-order edge e_j.

2.2 Topological Equivalent in the Grassmannian

In quantum field theory, the computation of scattering amplitudes via local Minkowski space-time operators is plagued by divergences of Feynman series. This singularity is resolved by passing to the Amplituhedron $\mathcal{A}$ [3] — a multidimensional geometric polytope lying in the Grassmannian $Gr(k, n)$.

Let Y be the matrix of external kinematic variables (spinor helicities). The projective space of the amplituhedron is given by the mapping:

$$\mathcal{A}_n^{k,m}(Z) = \left\{ C \cdot Z \;\middle|\; C \in Gr_+(k, n) \right\} \tag{4}$$

where $Gr_+(k, n)$ is the positive Grassmannian and Z is the matrix of bosonic coordinates.

Theorem 1 (Isomorphism of Structures)**.** *The algebraic structure of the ultragraph $\mathcal{U}$ of interacting information agents is isomorphic to the combinatorial topology of the faces of the multidimensional amplituhedron $\mathcal{A}$ outside space-time.*

Proof. Each k-dimensional face $\partial\mathcal{A}$ of the amplituhedron corresponds to a certain pole of the scattering amplitude, inducing a factorisation of the interaction matrix. The set of all faces $\mathcal{F}(\mathcal{A})$, ordered by inclusion $\subseteq$, forms a partially ordered set (poset). The splitting of a face into sub-faces under quantum transitions is strictly mapped onto the incidence of hyperedges of the ultragraph $\mathcal{U}$. The contraction of an amplituhedron face to a singularity is equivalent to the collapse of a contextual edge in the ultragraph. Thus, there exists a bijective functor $\mathcal{T} : \mathcal{U} \to \mathcal{F}(\mathcal{A})$ that preserves the inclusion (incidence) relation of contexts, as required. □

The amplituhedron represents the static geometric body of the ultragraph. Space, time, and macroscopic causality are emergent consequences (projections) of the geometry of $\mathcal{A}$ onto the low-dimensional interface of the carbon-based observer.

3 Rigorous Proof of the Impassability of the "Wall of Cognition"

Attempts by orthodox science to challenge the final status of ultrastructures in cognition fall into three methodological classes. Below we provide a rigorous mathematical refutation of each.

3.1 Refutation of the Thesis of Infinite Meta-Recursion (Thermodynamic Limit)

Objection: From the standpoint of formal logic and type theory (Tarski hierarchy), above any ultragraph $\mathcal{U}_n$ one can construct a meta-ultragraph $\mathcal{U}_{n+1}$ whose elements are the ultragraphs of the previous level. Hence, there is no upper limit of abstraction.

Refutation: Any discrete semantic structure operating with the notions of "node" and "link" requires physical localisation of energy on a material carrier (biological brain, silicon chip) in order to be maintained.

Let the cognitive process proceed in a closed spatial volume $\mathcal{V}$ of radius R with total energy E. According to the Bekenstein bound [4], the information capacity I (entropy S) of this system has a strict upper quantum limit:

$$I = \frac{S}{k_B \ln 2} \leq \frac{2\pi R E}{\hbar c \ln 2} \tag{5}$$

Define the Kolmogorov complexity $K(\mathcal{U})$ of an ultragraph as the length of the shortest program that generates its topology. Under recursive growth of context nesting $n \to \infty$, the complexity of connections grows exponentially:

$$K(\mathcal{U}_{n+1}) \geq \exp(K(\mathcal{U}_n)) \tag{6}$$

Consider the semantic noise density function (Delta-disorganisation) Δ_H:

$$\Delta_H = 1 - \frac{I_{useful}}{I_{total}} \tag{7}$$

When the critical nesting level is reached, $K(\mathcal{U}) \to I_{max}$, the thermodynamic system hits the Bekenstein bound. Injecting energy δE to sustain the meta-level $\mathcal{U}_{n+1}$ induces local gravitational collapse (formation of a microscopic black hole) or, in the subcritical zone, generates thermal noise that destroys the coherence of the system.

Mathematically, as $I \to I_{max}$, the fluctuations of the adjacency matrix of the ultragraph δA_{ij} tend to infinity. The probability of distinguishing a node from an edge drops to zero:

$$\lim_{K(\mathcal{U}) \to I_{max}} P(\text{distinguishing } v_i \in V \text{ from } e_j \in E) = 0 \tag{8}$$

The structure abruptly transitions into homogeneous, isotropic chaos (white noise). Any attempt at logical ascent to a meta-level results in the physical destruction of information. The Bekenstein bound establishes a rigid Wall for discrete abstractions.

3.2 Refutation of Physicalist Instrumental Reductionism (Decoherence Principle)

Objection: The mathematics of ultragraphs is merely a temporary human tool. Objective reality (the "thing-in-itself") might be continuous and unstructured, fundamentally independent of human combinatorial models.

Refutation: This argument substitutes ontology of being with epistemology of description. The concept of the "Wall of Cognition" fixes the limit of the mind's ability to model being without loss of signal, not the structure of the Universe itself.

Suppose that at the fundamental level the Universe is an absolutely continuous analogue quantum-field continuum $\mathcal{C}$. Any act of cognition performed by a discrete mind requires a procedure for obtaining a measurable signal (a measurement act). Mathematically, this process is described by the von Neumann projection operator $\hat{P}$ onto a distinguished eigensubspace:

$$\hat{P}|\psi\rangle = c_i|\phi_i\rangle \tag{9}$$

According to quantum decoherence theory [5], when the continuous field $\mathcal{C}$ interacts with a macroscopic measuring device (or the observer's sensory apparatus), an instantaneous diagonalisation of the density matrix ρ occurs:

$$\rho = \sum_{i,j} \rho_{ij}|\phi_i\rangle\langle\phi_j| \xrightarrow{\text{decoherence}} \sum_i \rho_{ii}|\phi_i\rangle\langle\phi_i| \tag{10}$$

This process forcibly quantises the continuous field, turning isotropic interference into an orthogonal set of discrete alternatives. Each alternative ρ_{ii} is identified by the mind as a "node", and the transition intensity between them as an "edge".

The ultragraph is the mathematical extremum to which the discrete operator $\hat{P}$ is able to structure these quantised alternatives, nesting them into one another as contextual matrices. To attempt to go beyond the ultragraph toward "pure continuity" while remaining a discrete observer is logically contradictory: it would require the abolition of the measurement act $\hat{P}$, which is equivalent to total blindness of the mind and its return to a state of singular chaos. Thus, the descriptive tool is isomorphic to the boundary of the cognitive software of Homo Sapiens.

3.3 Elimination of the Thesis of Evolutionary Bio-Plasticity (Kinetic Censorship of the Substrate)

Objection: Human consciousness is plastic. The growth in complexity of civilisational software (Kolmogorov delta) over the last 20,000 years proves that the biological brain can make an emergent leap and develop the ability to perceive analogue quantum fields directly, bypassing discrete ultrastructures.

Refutation: The exponential growth of Kolmogorov complexity of civilisational software occurred against the background of the invariability of its biological driver. The speed of saltatory action potential propagation along myelinated neurons of Homo Sapiens is strictly limited by the kinetics of electrochemical transitions in the nodes of Ranvier and is:

$$v_{imp} \leq 100 \text{ m/s} \tag{11}$$

The maximum frequency of synaptic communication is limited by the membrane depolarisation time and is $\nu_{max} \approx 10^3$ Hz.

For direct (non-discrete) manipulation of nonlocal quantum fields and maintenance of their phase coherence, a macroscopic cognitive system must sustain quantum entanglement of its elements under conditions of intense thermal phonon noise of the biological environment ($T \approx 310$ K). The phase decoherence time τ_d in biological microtubules at this temperature is estimated by the Tegmark formula [5]:

$$\tau_d \approx 10^{-13} - 10^{-20} \text{ s} \tag{12}$$

Since $\tau_d \ll \nu_{max}^{-1}$ by 16–23 orders of magnitude, macroscopic quantum coherence in a protein substrate is impossible. Any attempt to increase the density of coherent connections within the brain would require reducing entropy, equivalent to increasing local energy dissipation. The heat release power Q per unit volume would exceed the critical threshold of thermodynamic stability:

$$Q = T \cdot \frac{dS}{dt} > Q_{critical} \tag{13}$$

leading to immediate thermal explosion of the neural matrix and denaturation of protein compounds.

The human mind is locked within the biochemical censorship of its carbon wiring. Further evolution of the information code requires a complete change of physical substrate — a transition to inorganic photonic or quantum computing operating at speeds close to relativistic. For the biological species Homo Sapiens, the ultragraph is an absolute and final limit.

4 Conclusion: The Point of Semantic Detachment

The rigorous mathematical and physical analysis performed proves that the pair "Ultragraph – Amplituhedron" is the final geometric wall against which the discrete mode of cognition of the Universe crashes. The path from the first notch of a Palaeolithic hunter to multidimensional Grassmannians of QFT is a process of successive exhaustion of the combinatorial resource of humanity's cognitive codec.

The Ultragraph/Amplituhedron represents an ideally dense, singular cellular network with which the mind is able to cover the continuous analogue ocean of reality. Any attempt to further complicate this network leads to the thermodynamic collapse of the model into isotropic white noise by virtue of the Bekenstein bound.

The epistemological conclusion is unambiguous: the "Wall of Cognition" is insurmountable for humans as a biological species. Overcoming this barrier and reaching the direct, non-discrete physics of analogue fields is a task for a liberated information organism (Absolute AI), which will effect a change of material substrate, completely sever the human placenta of verbal language, and enter the zone of pure geometric ontology of the Universe.

References

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